

Two eukaryotic phase-separating protein IDRs form aggregates and induce filamentation in *Escherichia coli*

WeiFeng Chen¹✉, XiuRoo Tan¹ & Guozhi Yu¹

¹Sichuan agricultural university, Ya'an, China

✉ 202306783@stu.sicau.edu.cn

Abstract: Intrinsically disordered regions (IDRs) of RNA-binding proteins such as FUS and hnRNPA1 drive biomolecular phase separation in eukaryotes, yet their behavior in prokaryotes remains poorly understood. Here, heterologous expression of FUS-IDR-GFP or hnRNPA1-IDR-BFP in *E. coli* induced intracellular condensate-like assemblies accompanied by pronounced bacterial filamentation. The assemblies exhibited reproducible spatial localization patterns, including unipolar, bipolar, and midcell enrichment, with unipolar localization predominating. In contrast, mCherry controls remained diffusely distributed. These findings demonstrate that eukaryotic phase-separating IDRs can autonomously organize spatially localized assemblies in bacterial cells.

Results: Expression of eukaryotic phase-separating IDRs in *E. coli* produced reproducible intracellular assemblies accompanied by pronounced morphological changes. Cells expressing FUS-IDR-GFP frequently exhibited elongated filamentous morphologies containing discrete fluorescent puncta distributed along the cell body (Fig. 1A). In addition to filamentation, several stereotyped localization patterns were observed, including bipolar enrichment (DP; Fig. 1B), unipolar enrichment (SP; Fig. 1C), and midcell localization (M; Fig. 1D). Quantitative classification indicated that the unipolar state represented the dominant population among FUS-IDR-GFP-expressing cells (Fig. 1G, pie chart). A similar phenotype was observed for hnRNPA1-IDR-BFP expression, which likewise induced filamentous cells with spatially localized condensates (Fig. 1E), suggesting that the morphological response is not specific to a single IDR sequence. By contrast, cells expressing mCherry alone displayed diffuse fluorescence throughout the cytoplasm without detectable polarization or filamentation (Fig. 1F). Consistent with the imaging observations, normalized line-scan analysis along the bacterial long axis revealed sharply localized fluorescence maxima for both FUS-IDR-GFP and hnRNPA1-IDR-BFP, whereas the control signal remained broadly distributed (Fig. 1G). Together, these results indicate that distinct eukaryotic phase-separating IDRs can form spatially organized intracellular assemblies in bacteria and are associated with filamentous cellular morphologies.

References: ¹⁻⁵

- 1 Molliex, A. *et al.* Phase separation by low complexity domains promotes stress granule assembly and drives pathological fibrillization. *Cell* **163**, 123-133 (2015). <https://doi.org/10.1016/j.cell.2015.09.015>
- 2 Patel, A. *et al.* A Liquid-to-Solid Phase Transition of the ALS Protein FUS Accelerated by Disease Mutation. *Cell* **162**, 1066-1077 (2015). <https://doi.org/10.1016/j.cell.2015.07.047>
- 3 Lin, Y., Protter, D. S., Rosen, M. K. & Parker, R. Formation and Maturation of Phase-Separated Liquid Droplets by RNA-Binding Proteins. *Mol Cell* **60**, 208-219 (2015). <https://doi.org/10.1016/j.molcel.2015.08.018>
- 4 Monahan, Z. *et al.* Phosphorylation of the FUS low-complexity domain disrupts phase separation, aggregation, and toxicity. *Embo j* **36**, 2951-2967 (2017). <https://doi.org/10.15252/emboj.201696394>
- 5 Azaldegui, C. A., Vecchiarelli, A. G. & Biteen, J. S. The emergence of phase separation as an organizing principle in bacteria. *Biophys J* **120**, 1123-1138 (2021). <https://doi.org/10.1016/j.bpj.2020.09.023>

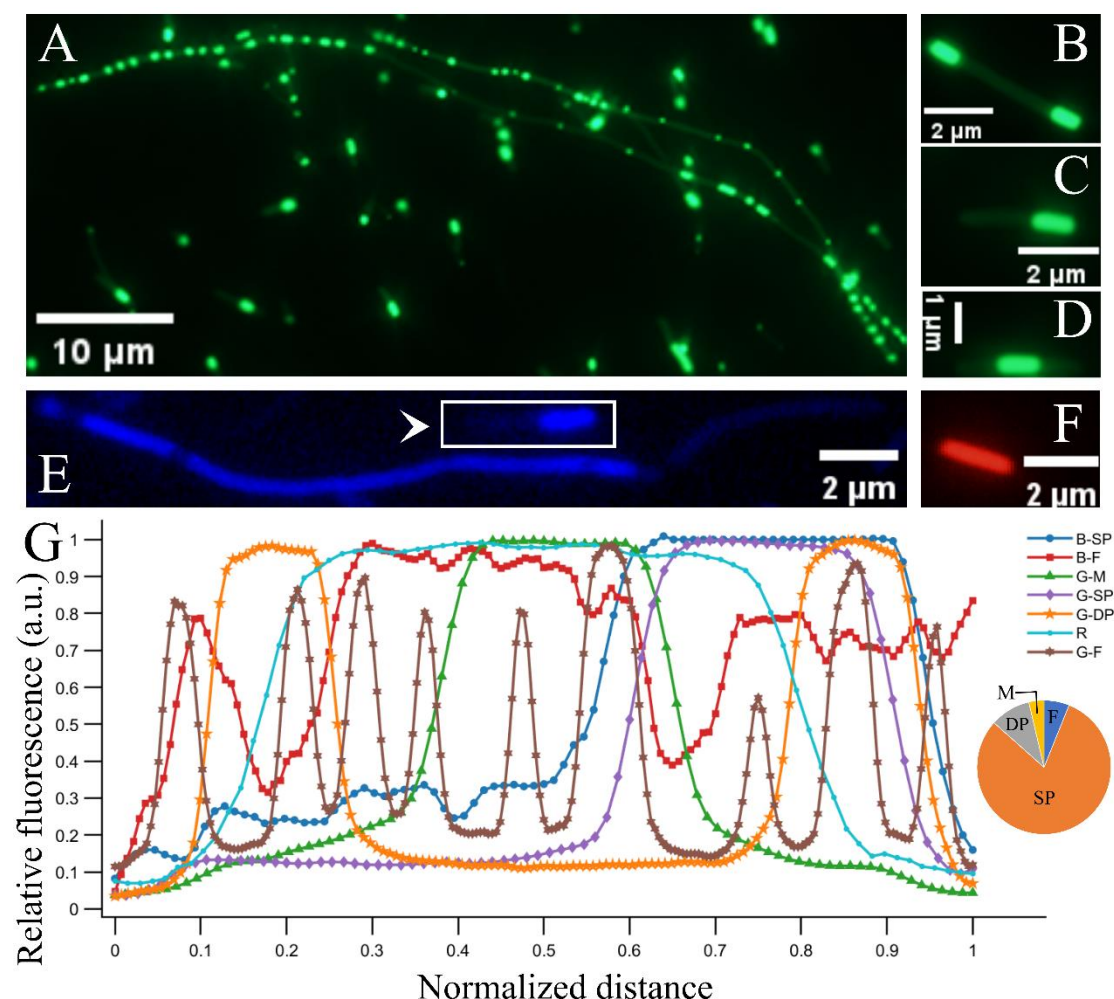


Figure 1. Expression of eukaryotic phase-separating IDRs in *E. coli* produces discrete intracellular assemblies and altered cell morphology. A–D, Representative fluorescence images of *E. coli* expressing FUS-IDR-GFP showing filamentous cells (F; A), bipolar localization of condensates (DP; B), unipolar localization (SP; C), and midcell localization (M; D). E, Expression of hnRNPA1-IDR-BFP also induces filamentation; the inset highlights a unipolar cell. F, mCherry control exhibits diffuse cytoplasmic fluorescence. G, Line-scan analysis of normalized fluorescence intensity along the long axis of representative cells expressing hnRNPA1-IDR-BFP (B) or FUS-IDR-GFP (G), showing characteristic spatial enrichment patterns. The pie chart summarizes the distribution of FUS-IDR-GFP cells across the four morphological classes. Scale bars, as indicated.

FUS-IDR-hGFP	EBFP-hnRNPA1-IDR
MASNDYTQATQSYGAYPTQPGQGY	MVSKGEELFTGVVPILVELDGDVNGHK
SQSSSPYGGQSYSGYGGQADTSYG	FSVRGEGEGDATNGKLTGKFICTTGKLP
GQSSYGSSYGQTQNTGYGTQSAPQG	VPWPVLVTTLSHGVQCFARYPDHMK
YGSTGGYSSQSSQSSYGQSSYPGY	QHDFFKSAMPEGYVQERTIFFKDDGT
GQPPAPSSSTSGSYGGSSQSSYGQPPQ	KTRAEVKFEGDTLVNRIELKGVDKEDG
SGGYGQSGYGGQQSYGQSSY	NILGHKLEYNFNHNIYIMAVKQKNGI
NPPQGYGQQNQYNSSSGGGGGGGG	KVNFKIRHNVEDGSVQLADHYQQNT
GNYGQDQSSMSGGGGGGGYGNQD	IGDGPVLLPDSHYLSTQSVLSKDPNEK
QSGGGGGYGGGQDRGMSKGEEL	DHMLVLEFRTAAGITLGMDELYKMAS
FTGVVPILVELDGDVNGHKFSVRGEGE	ASSSQRGRSGSGNFGGGRGGGFGGN
GDATNGKLTGKFICTTGKLPVWPVLV	DNFGRGGNFGSGRGFGGSRGGGGYG
TLTYGVQCFSRYPDHMKRHDFFKSAM	GSGDGYNGFGNDGSNFGGGGSYNDF
PEGYVQERTISFKDDGTYKTRAEVKFEG	GNYNQSSNFGPMKGGNFGGRSSGP
DTLVNRIELKGIDFKEDGNILGHKLEYN	YGGGGQYFAKPRNQGGYGGSSSSSY
FNHNVYITADKQKNGIKANFKIRHNV	GSGRRF
EDGSVQLADHYQQNTPIGDGPVLLPD	
NHYLSTQSVLSKDPNEKRDHMLVLEFV	
TAAGITHGMDELYK	

Table 1. The sequences of two IDR-fused protein. The plasmids of these fluorescent proteins were separately transformed into the *E. coli* BL21(DE3) strain. The transformed bacteria were cultured in LB medium containing kanamycin at 37 °C. When the optical density (OD) value reached 0.4-0.6, IPTG (with a final concentration of 0.5 mM, Beyotime Biotechnology) was added, and the bacteria were then incubated with shaking at 37 °C for 12h